

AstraPay Payment Profitability

Roshan J Singh

May contribution profit

fell 85%

while attempted payment value rose 0.4%

49,980 payment attempts | January to June 2025

Data model

One row per payment attempt, with child tables aggregated before joining

AstraPay Payment Profitability

Entity relationship diagram • Primary keys, foreign keys and analysis joins

MASTER AND REFERENCE DATA

CUSTOMER	
One row per customer	
PK customer_id	STRING
segment	STRING
country	STRING
onboarding_date	DATE
kyc_status	STRING

MERCHANT	
One row per merchant	
PK merchant_id	STRING
category	STRING
country	STRING
pricing_plan	STRING
risk_tier	STRING

ROUTE_COST	
One row per payment route	
PK route_id	STRING
provider	STRING
region	STRING
fixed_fee	DECIMAL
variable_fee_pct	DECIMAL

FX_RATE	
One row per date, currency and rate type	
PK rate_date	DATE
currency	STRING • PK
rate_to_usd	DECIMAL
rate_type	STRING • PK
Composite PK: rate_date + currency + rate_type	

MERCHANT_RISK_SNAPSHOT	
One row per merchant and month	
PK FK merchant_id	STRING
snapshot_month	DATE • PK
risk_score	DECIMAL
risk_band	STRING
Expected PK: merchant_id + snapshot_month	

CENTRAL PAYMENT FACT

TRANSACTION_DATA	
Central fact table • One row per payment attempt	
PK transaction_id	STRING
FK customer_id	STRING
FK merchant_id	STRING
amount	DECIMAL
currency	STRING
channel	STRING
status	STRING
created_at	TIMESTAMP
FK route_id	STRING

TRANSACTION DETAIL AND OUTCOME DATA

PAYMENT_EVENT	
Multiple lifecycle events per transaction	
PK FK transaction_id	STRING
event_type	STRING • PK
event_time	TIMESTAMP • PK
processing_ms	INTEGER
ingestion_time	TIMESTAMP
Composite key: transaction + event type + time	

FRAUD_DECISION	
Fraud model and rule decisions	
PK FK transaction_id	STRING
rule_id	STRING • PK
decision	STRING
risk_score	DECIMAL
model_version	STRING • PK
Composite key: transaction + rule + model version	

SETTLEMENT	
Zero or more records per transaction	
FK transaction_id	STRING
settlement_amount	DECIMAL
settlement_currency	STRING
fee_amount	DECIMAL
settlement_status	STRING
settlement_date	DATE
No reliable unique primary key in source	

CHARGEBACK	
Customer disputes linked to transactions	
PK chargeback_id	STRING
FK transaction_id	STRING
reason	STRING
amount	DECIMAL
opened_at	TIMESTAMP
resolved_at	TIMESTAMP

Data controls preserve the May decline signal

Data quality exceptions

Data condition	Records	Rate
Missing merchant ID	401	0.8%
Zero or negative amount	412	0.8%
Invalid fraud score	506	1.0%
Late payment event	4,175	8.4%
Missing settlement	8,465	16.9%
Duplicate risk snapshot	496	1.0%

Treatment decisions

Issue	Decision
Missing merchant ID	Retain and label Unknown
Invalid fraud score	Retain payment; exclude score
Missing settlement	Keep attempt; measure coverage
Duplicate risk snapshot	Deduplicate merchant and month

Settlement coverage stayed near 92% in both April and May. The trusted model preserves the direction of the decline.

Blocking controls focus on grain, currency and temporal joins.

Profit per successful payment fell from \$0.70 to \$0.11

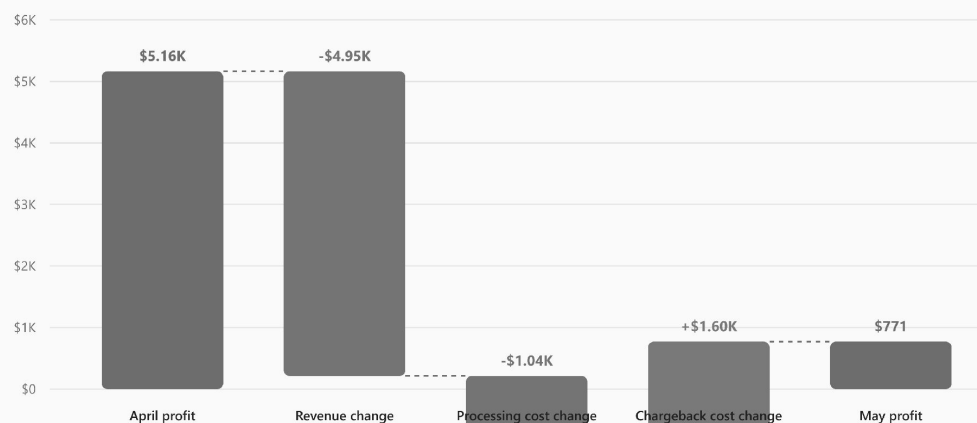
April to May 2025

Contribution profit	Metric comparison			
	Metric	April	May	Change
Merchant fee revenue				
less processing cost	Payment attempts	8,268	8,578	+3.7%
less observed chargeback cost	Success rate	89.6%	85.3%	-4.3 pp
Calculated once at transaction grain	Attempted value	\$2.27M	\$2.28M	+0.4%
	Revenue	\$30.15K	\$25.20K	-\$4.95K
	Processing cost	\$15.19K	\$16.23K	+\$1.04K
	Contribution profit	\$5.16K	\$0.77K	-\$4.39K
	Profit margin	17.1%	3.1%	-14.1 pp

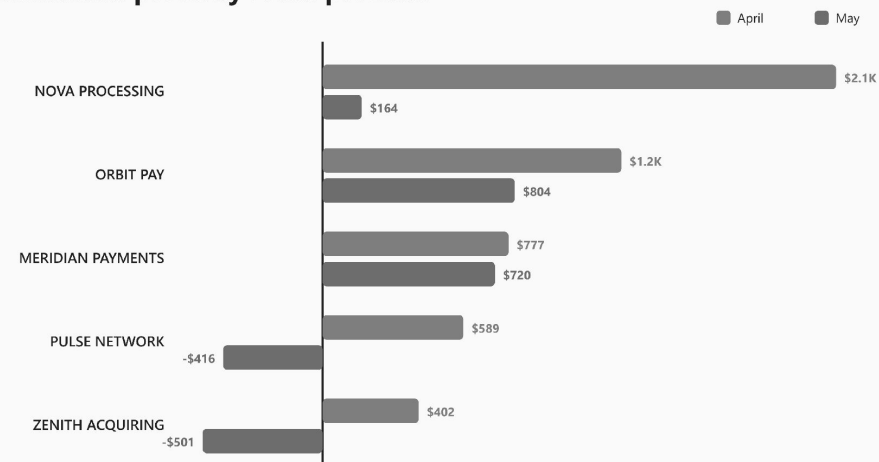
Attempted value increased, but fewer payments succeeded and each successful payment produced less profit.

Within-segment performance drove most of the May decline

April to May profit bridge



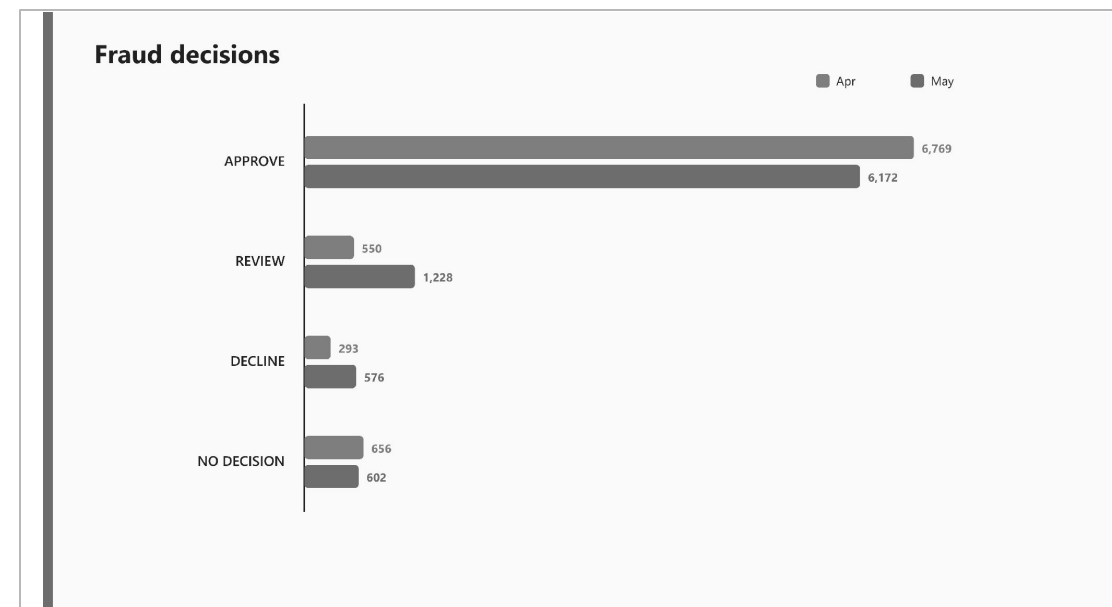
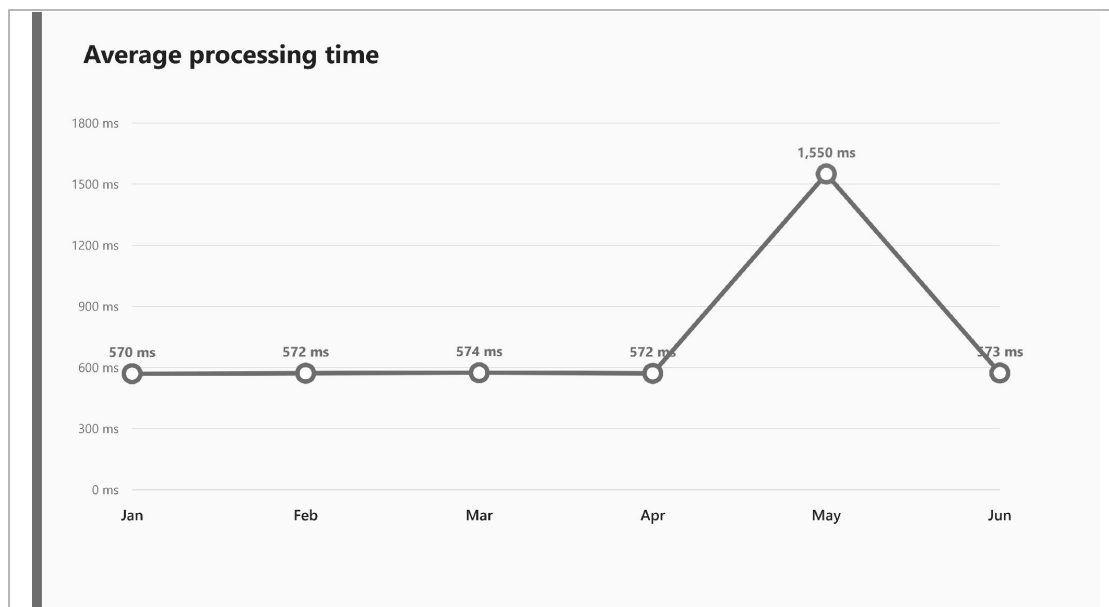
Contribution profit by route provider



Revenue fell \$4.95K. The \$1.04K processing-cost increase added pressure, while lower chargeback cost offset \$1.60K.

Nova, Pulse and Zenith accounted for \$3.89K of segment deterioration. Across the five grouping views, weak performance within segments outweighed mix changes.

May combined operational and commercial deterioration



Driver	April	May	Judgement
Average processing time	572 ms	1,550 ms	Strong operational deterioration
Fee yield on GPV	1.47%	1.28%	Material revenue pressure
Review and decline rate	10.20%	21.03%	Tighter fraud controls are plausible
FX conversion variance	-\$2.56K	-\$1.89K	Improved, so not a cause
Chargeback cost	\$9.80K	\$8.20K	Fell, so not the main cause

60-day action plan

Routing cost contributed, but the evidence weakens the claim that it was the main cause

Owner	Action	Expected effect	Main risk	Success measure	Monitoring data
Payments Ops	Fix May latency by route and provider	Restore approval and completion	Merchant disruption	Average below 700 ms and success above 89%	Events and route logs
Finance and Pricing	Review fee yield for Retail, Singapore, Web and Mobile	Recover about 19 bps of yield	Merchant churn	Fee yield at or above 1.45%	Pricing and settlement fees
Risk	Backtest May reviews and declines	Reduce avoidable declines	Higher fraud loss	Review and decline at or below 12% with stable chargebacks	Decision outcomes and chargebacks
Data Engineering	Add route dates and enforce safe grain	Trusted and repeatable reporting	Backfill delays	No amount inflation and reconciliation variance below 1%	Route history and settlements
Analytics	Monitor the governed view each week	Faster action on deterioration	Metric drift	Signed weekly KPI review	Trusted profitability model
Decision	Treat May as a combined revenue, fraud-control and processing problem. Validate fraud outcomes before relaxing controls.				